Md Shamimul Islam

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Research Interests

Computer Vision | Medical Imaging | Human Activity Recognition | Multimodal Learning

Education

2025 – Present
- **MS in Computer Science, Florida Polytechnic University**
  CGPA: 3.77 out of 4.00
  Relevant coursework: Machine Learning and Practice (A), Scientific Computation and Programming (A), Advanced Algorithm Design and Analysis (A).
  In progress (Fall 2026): Intelligent Systems, Thesis I.

2020 – 2022
- **Masters in Applied Statistics and Data Science, Jahangirnagar University (JU)**
  CGPA: 3.96 out of 4.00 (ranked 1st in batch)
  Thesis title: *Efficient Deep Learning Methods for Continuous Bangla Sign Language Recognition.*
  Relevant coursework: Machine Learning for Data Science (A+), Probability and Probability Distributions (A+), Statistical Inference (A+), Multivariate Analysis (A+), Applied Regression Analysis (A+), Time Series Analysis and Forecasting (A+), Data Mining (A), Introduction to Data Science with Python (A+).

2015 – 2020
- **B.Sc. in Computer Science and Engineering, Manarat International University (MIU)**
  CGPA: 3.99 out of 4.00 (ranked 1st in batch)
  Thesis title: *Pose-based Multi-level Features Fusion For Exam Cheat Recognition Using CNN.*
  Relevant coursework: Artificial Intelligence, Computer Vision and Robotics, Neural Network and Fuzzy Systems, Digital Signal and Image Processing, Linear Algebra and Matrix, Statistics and Probability, Numerical Methods (all A+).

Research Publications

Peer-Reviewed Publications (Total Citation: 105 | h-index: 4 | i10-index: 2)

1. Al-Mahmud*, M. S. Islam*, and M. N. Hossain, "Video-based sexual harassment detection using fine-tuned video transformers," *Co-first authors; contributed equally. Accepted to Proceedings of the IEEE 3rd International Conference on Computing, Applications and Systems, University of Dhaka, Bangladesh, 2026.

2. **M. S. Islam**, L. G. Jaimes, and A. S. Dina, "MA-IDS: Multi-agent RAG framework for IoT network intrusion detection with an experience library," in *Proceedings of the 2026 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, Accepted for publication, Bellevue, WA, USA, 2026. arXiv: 2604.05458 [cs.CR], (Citations: 6).

3. S. Jannat, M. M. Hasan, **M. S. Islam**, and M. N. Hossain, "Bat-sam: Introducing inductive bias into sam via a mixture of convolutional experts for brain tumor segmentation," Accepted to 12th IEEE Int. Women in Engineering (WIE) Conf. on Electrical and Computer Engineering, 2026., 2026.

4. **M. S. Islam**, A. J. M. A. Joha, N. Hossain, M. M. Hasan, and S. Abdullah, "Multi-view bangla sign language recognition: A new word-level video dataset," in *2024 IEEE The 6th International Conference on Activity and Behavior Computing*, Nakatsu and Kitakyushu, Kyushu, Japan, 2024. ↗DOI: 10.1109/ABC61795.2024.10652018 (Citations: 7).

5. M. M. Hasan, **M. S. Islam**, and S. Abdullah, "Robust pose-based human fall detection using recurrent neural network," in *IEEE International Conference on Robotics, Automation, Artificial-Intelligence and Internet-of-Things*, Dhaka, Bangladesh, 2019. ↗DOI: 10.1109/RAAICON48939.2019.23 (Citations: 62).

6. **M. S. Islam**, M. M. Hasan, and S. Abdullah, "A deep spatio-temporal network for vision-based sexual harassment detection," in *IEEE International Conference on Emerging Technology in Computing, Communication and Electronics*, 2021. ↗DOI: 10.1109/ETCCE54784.2021.9689891 (Citations: 22).

7. **M. S. Islam** and I. Elwarfalli, "Deep learning-powered ecg-based biometric authentication," in *IEEE International Conference on Next-Generation Computing, IoT and Machine Learning (NCIM)*, DUET, Dhaka, Bangladesh, 2023. ↗DOI: 10.1109/NCIM59001.2023.10212845 (Citations: 4).

8. N. H. M. Arafat, S. Ileas, Mujazid, and **M. S. Islam**, "Blosecr: A conceptual framework of identity documents' security on blockchain in refugee issues and conserving fundamental rights," in *The International Conference for Intelligent Technologies*, Karnataka, India, 2022. ↗DOI: 10.1109/CONIT55038.2022.9847731 (Citations: 4).

Under Review / In Preparation

1. **M. S. Islam** and A. S. Dina, "Sera-ids: Structured experience retrieval-augmented intrusion detection with small language models," Submmitted to IEEE International Conference on Communications (ICC) 2027, 2026.

2. **M. S. Islam**, A. H. Sohag, and Z. Subah, "Permissionguard: Fine-tuned LLMs and BERT for android permission-based malware detection," Manuscript in preparation, 2025.

Datasets

1. A. J. M. A. Joha, M. N. Hossain, M. S. Islam, S. Abdullah, and M. M. Hasan, *Multi-View Bangla Sign Language (MV-BSL) Dataset*, GitHub repository, 2024. ↗URL: https://github.com/Mahedi-61/MV-BSDL

2. S. Abdullah, M. S. Islam, and M. M. Hasan, *Bangladeshi Vehicle License Plate Dataset*, GitHub repository, https://github.com/Mahedi-61/Bangla_License_Plate_Dataset.

3. M. S. Islam and M. M. Hasan, *Sexual Harassment Video Dataset (SHVD)*, GitHub repository, https://github.com/Shamim-38/Sexual-Harassment-Video-Dataset-SHVD-.

Employment History

2025 – ···· • **GA,** Department of Computer Science, Florida Polytechnic University.

2024 – 2025 • **Lecturer,** Department of Computer Science and Engineering (CSE), Manarat International University (MIU).

Employment History (continued)

2023 – 2024
- **Lecturer,** Department of CSE, Asian University of Bangladesh (AUB)
  Convener, AUB Programming Club.

2020 – 2022
- **Adjunct Faculty,** Department of CSE, MIU.

Research Projects

2025 – Present
- **Experience-Augmented LLM/SLM Frameworks for Network Intrusion Detection (Thesis):** Developed MA-IDS and SERA-IDS, RAG-based frameworks that learn reusable intrusion-detection knowledge from classification errors. MA-IDS uses GPT-4o with a FAISS Experience Library (IEEE SMC 2026), while SERA-IDS extends it with structured, confidence-gated retrieval for locally deployable SLMs without fine-tuning (submitted to IEEE ICC 2027).
- **Video-Based Sexual Harassment Detection Using Video Transformers:** Fine-tuned TimeSformer, VideoMAE, and ViViT for video-based harassment detection. TimeSformer achieved 92.27% test accuracy. Accepted at IEEE 3Cs 2026.

2026 – Present
- **PermissionGuard: LLM-Based Android Malware Detection:** Compared LoRA-tuned LLaMA 3.1-8B and BERT with classical ML models for permission-based malware detection, achieving up to 97.00% accuracy. Manuscript in preparation.

2024 – 2025
- **AI Career Counselor & Guidance Platform:** Built a LLaMA-powered platform for personalized IT career recommendations using an 8,000-sample dataset.

2024
- **Phishing URL Detection:** Developed an ML-based phishing detection system using Logistic Regression, Random Forest, SVM, and Neural Networks.

2022 – 2023
- **Real-Time Bangla Sign Language Translation:** Developed a Transformer-based system for translating pose-based sign-language video representations into sentences; winner of AI for Bangla 2.0.

2022
- **Bangla Social Network Post Analysis:** Developed a Transformer-based model for political sentiment and toxicity classification of Bangla social-media posts.

2018 – 2019
- **Automatic License Plate Recognition (ALPR):** Developed a real-time ALPR system using a custom Bangladeshi license-plate dataset; published at an IEEE conference.

Miscellaneous Experience

Awards and Achievements

2023
- **Winner of AI for Bangla 2.0 Competition:** Led a team to victory in the national-level competition directed by the Bangladesh Government, developing an innovative AI solution for Bangla sign language among 100+ competing teams.

2020
- **Islamic Development Bank (IsDB) Masters Scholarship:** Awarded the highly competitive scholarship ($6,000) for outstanding academic performance in graduate studies.

Miscellaneous Experience (continued)

2018 • **University Merit Scholarship:** Awarded a full tuition waiver for placing in the top 1% of undergraduate students at Manarat International University.

Certification

2015 • **Champion,** Great Victory Day Essay Competition, Manarat International University.

Co/Extracurricular Activities

2026–Present • **Vice President,** Muslim Students Association (MSA), Florida Polytechnic University, Lakeland, FL.

• **Research/Workshop Coordinator,** AI, Robotics, and Vision Research Club (AIRVRC), Florida Polytechnic University, Lakeland, FL.

2026 • **Volunteer,** Summer Camp 2026, SALAM Academy Florida, FL. Assisted with community activities and student group management.

2019-2020 • **Elected President,** MIU CSE Club, Manarat International University, Ashulia, Bangladesh. Spearheaded department-wide initiatives and organized major events including MIU CSE Fest 2020 and the Intra-University Programming Contest.

• **Organizing Member,** Code Camp, MIU CSE Club, Manarat International University, Ashulia, Bangladesh.

2019 • **Participant,** GDG Sonargaon Hackathon, Google Developer Group, Dhaka, Bangladesh.

2018 • **Participant,** ICPC Bangladesh National Collaborative Programming Contest (NCPC), IUBAT, Dhaka, Bangladesh.

2017 • **Co-founder,** Hasimukh (Smile), Rangpur, Bangladesh. Established a social welfare organization dedicated to supporting underprivileged communities.

Talks and Poster Presentations

2026 • **Invited Speaker,** 2026 Engineering and Science Student Research Symposium, Florida Polytechnic University, Lakeland, FL. Talk: "Adaptive Reasoning for Network Traffic Security: A Multi-Agent RAG Architecture with Persistent Knowledge Induction." Advisor: Dr. Ayesha S. Dina.

• **Poster Presentation,** "MA-IDS: Multi-Agent RAG Framework for IoT Network Intrusion Detection with an Experience Library," Florida Polytechnic University, Lakeland, FL.

2022 • **Invited Speaker,** Webinar on "A Publication Road Map for Scientific Papers," organized by MIU CSE Club, MIU, Ashulia, Bangladesh.

Skills

Languages	• Strong reading, writing and speaking competencies for Bengali and English.
Coding	• Python, Java, PHP, C, R, SQL, XML/XSL.
System Administration	• Unix/Linux, Windows
Document Processing	• LaTeX, MS Office
Developer Tools	• Git, Google Cloud Platform
Libraries	• NumPy, OpenCV, Pandas, Matplotlib
AI Framework	• PyTorch, TensorFlow

Skills (continued)

Object Detection Algorithms	• YOLO, RetinaNet, Mask R-CNN
Pose Detection Algorithm	• OpenPose, MediaPipe
Misc.	• Academic research, teaching, training, consultation, LaTeX typesetting and publishing.

References

Available upon request.